Perimeters and areas of similar figures

1

a $18\ \text{cm}^2$ **b** $72\ \text{cm}^2$ **c** $1 : 4$ **d** Yes

2

a $\frac{3}{7}$ **b** $\frac{9}{49}$

3 12

4

a 36 **b** $504\ \text{cm}^2$

5

a $81 : 169$ **b** $169\ \text{cm}^2$

6

a 3

b $21\ \text{cm}, 30\ \text{cm}, 48\ \text{cm}$

c The perimeter of the larger triangle is 3 times larger than the perimeter of the smaller triangle.

7 $324\ \text{cm}^2$

8 $49 : 4$

9 It is 9 times the original area.

10

a $128\ \text{cm}^2$ **b** $9\ \text{cm}$

11 $600\ \text{cm}^2$

12 $\frac{25}{16}$

13

a 4 **b** 16 **c** 24

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14

a A will increase by a factor of 4 in total.

b A will remain unchanged.

15

a A will increase by a factor of 9.

b A will decrease by a factor of 16.

Additional questions

16 $\frac{36}{25}$ times bigger

17 $6\ \text{m}^2$

18 The enlarged pentagon has a perimeter that is 3.25 times greater than the original pentagon.

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